MICAI 2026

TUTORIAL · MICAI 2026

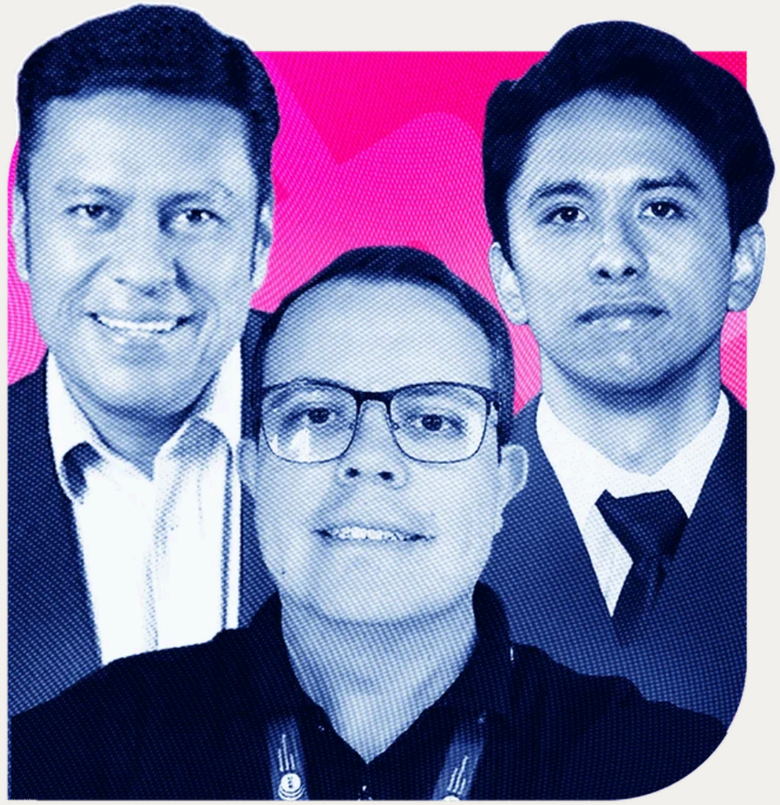
Opening the Black Box

A Hands-On Introduction to Mechanistic Interpretability of Large Language Models

Tec de Monterrey, Campus Chihuahua · Friday 6 November 2026

Miguel González · **Salvador Hinojosa** · **Víctor Terrón-Macías**

Tecnológico de Monterrey



THIS HOUR

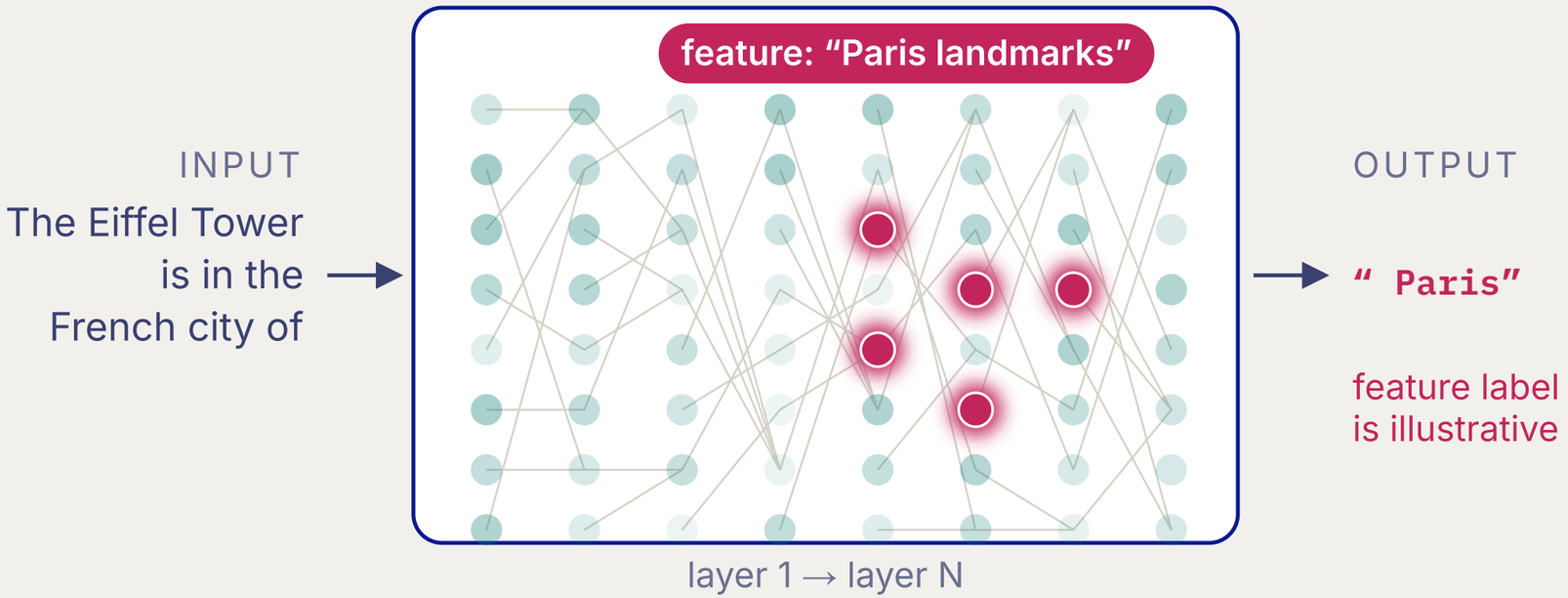
Four blocks compressed, the fifth in full

TIME	BLOCK	WHAT WE DO
0:00	**1 → 4 · speed-run**	Three ideas only: directions carry meaning, one stream to read them, intervention tests them
0:09	**5 · Audit**	Can we trust an LLM judge? A probe, a grounding check, a threshold
0:28	**NB5 · results**	What the notebook measures, and what we measured — walked through, not run
0:39	**Debrief**	What an auditor would accept, and the method map
0:45	**Wrap**	Where the field is going · open problems · resources

> Four hours and five Colab notebooks in the full version. Today: the results.

Full tutorial at MICAI 2026 · 6 November 2026 · with Miguel González and Salvador Hinojosa.

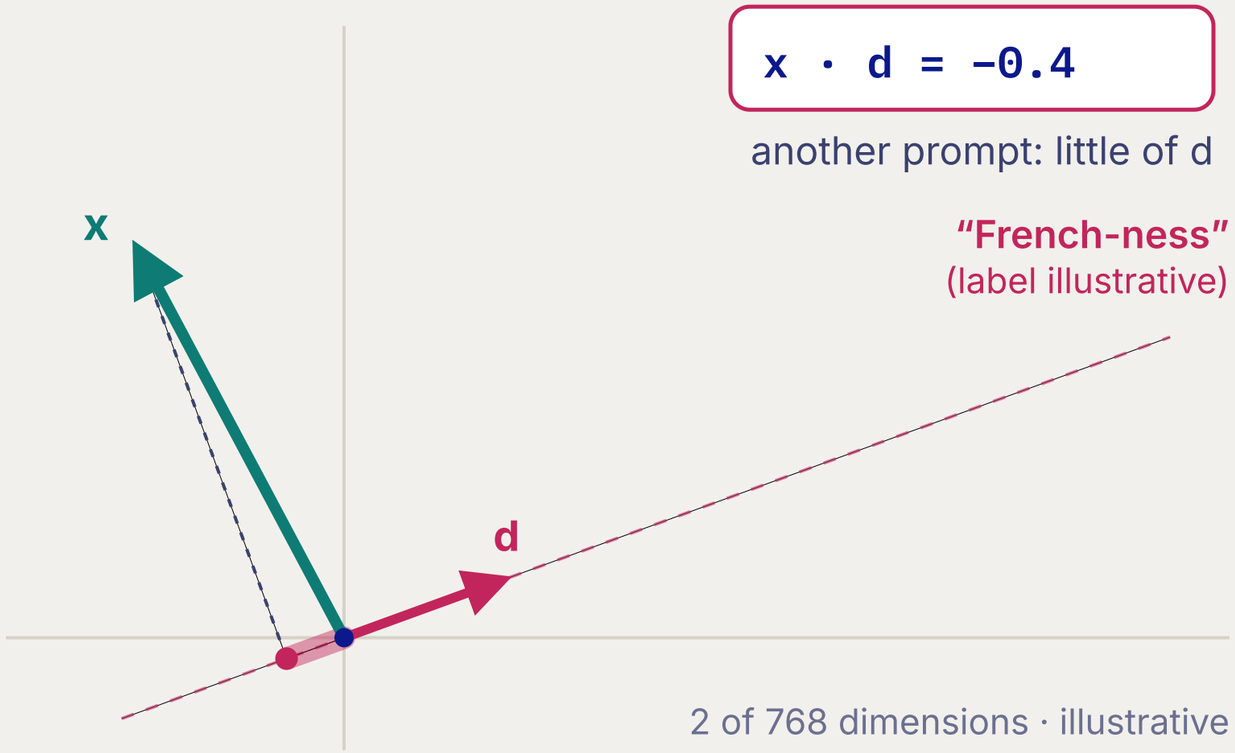
Tests see outputs; interpretability reads the computation



Testing: inputs in, outputs out.

Interpretability: read the steps in between.

A direction in activation space can mean something



The same move, all day:

Logit lens — Block 1

`x · (a token's vector)`

Probe — Blocks 3, 5

`x · w, w learned`

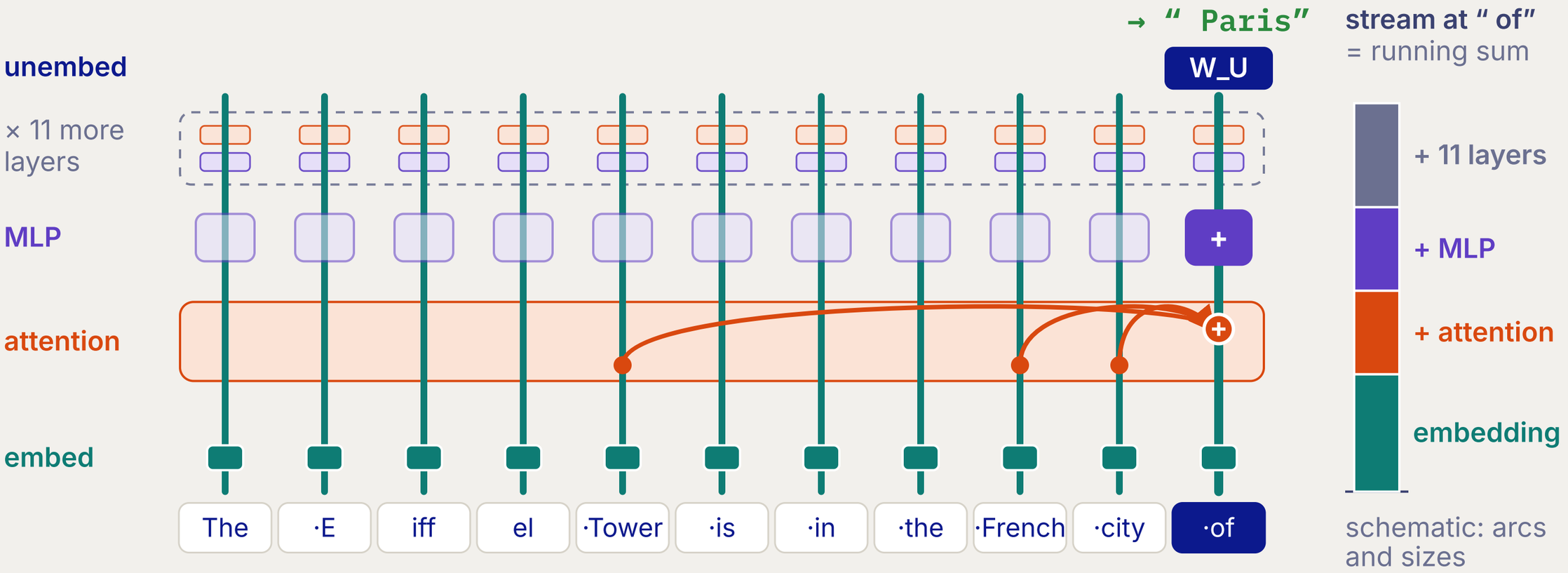
SAE feature — Blocks 3, 4

$x \cdot w_{enc}$, then (Jump)ReLU

Steering — Block 3

`x + α·d: writes along d`

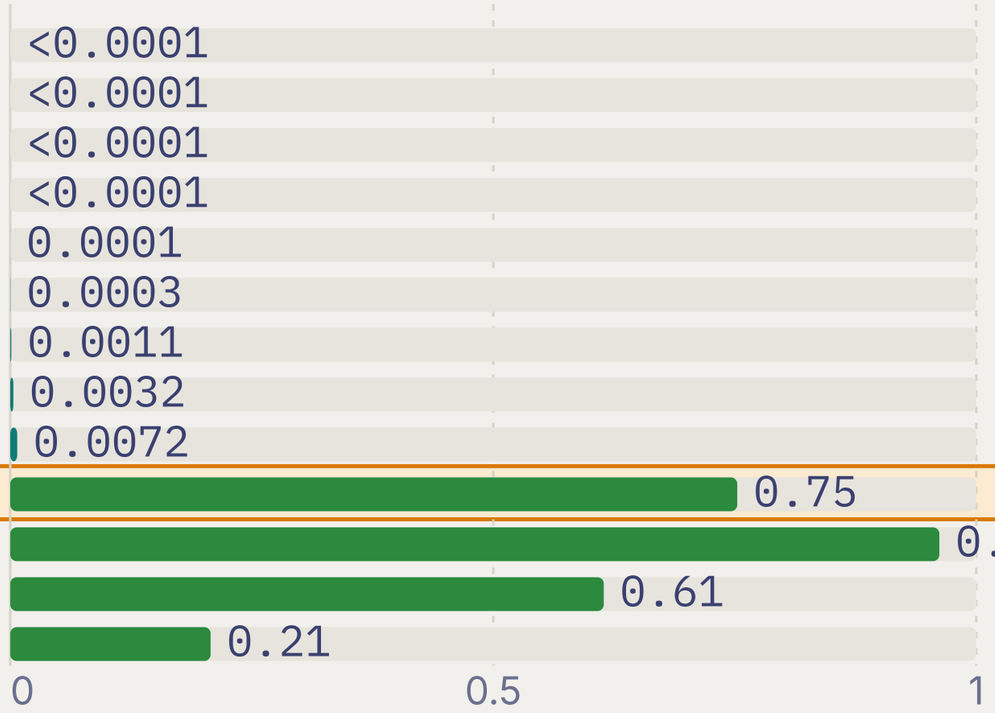
Every layer reads from one shared stream and adds to it



After 12 layers, **unembedding** reads the final stream at “ of” → next token “ Paris”.

Decoding every layer shows “Paris” emerging late

RESIDUAL AFTER	TOP-1 TOKEN	P(·PARIS)	RANK OF ·PARIS
embed	·destro	<0.0001	23,918
block 0	·the	<0.0001	5,323
block 1	·the	<0.0001	4,093
block 2	·the	<0.0001	1,804
block 3	·the	0.0001	374
block 4	·the	0.0003	233
block 5	·the	0.0011	128
block 6	·La	0.0032	46
block 7	·Man	0.0072	21
block 8	·Paris	0.75	1
block 9	·Paris	0.96	1
block 10	·Paris	0.61	1
block 11	·Paris	0.21	1



“ Paris” is top-1 from **block 8** on (after 9 of 12 blocks). Real GPT-2 small values.

HONEST LIMITS

Observation shows correlation, not cause

The logit lens can tell us

By block 8, the stream already points toward “ Paris”.

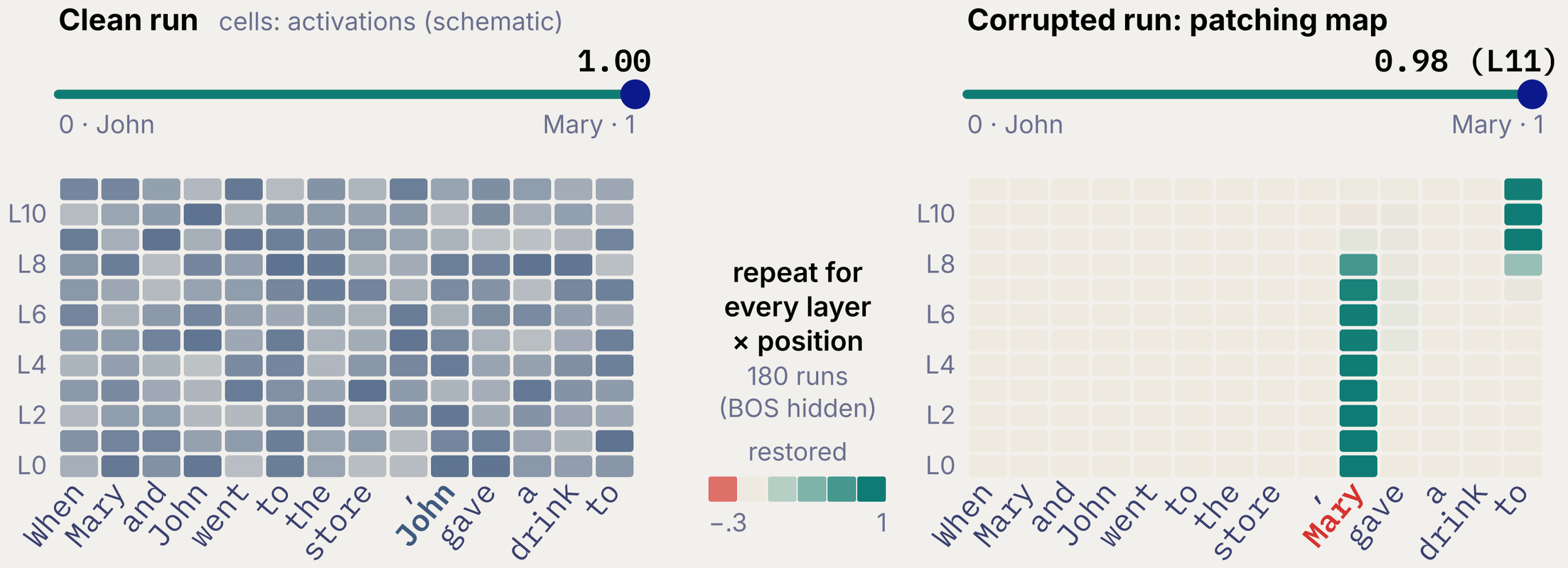
It cannot tell us

That block 8 *causes* the answer, or which component computes it.

- Attention patterns show where a head looks, not what it moves or whether it matters.
- Early layers decode poorly: their basis drifts from the output's. The tuned lens learns a translator per layer.

nostalgebraist (2020) “interpreting GPT: the logit lens”. Belrose et al. (2023) “Eliciting Latent Predictions from Transformers with the Tuned Lens”. Jain & Wallace (2019) “Attention is not Explanation”.

A patch transplants one activation between runs



Each method answers one question and leaves one open

METHOD	ANSWERS	LEAVES OPEN
Patching	where the model relies on information	what that information is
SAEs · probes · steering	which units are present; can we detect them	whether they are *the* variables
Attribution graphs	how units combine on one prompt	whether it generalizes

> The common rule: an explanation counts when an intervention can test it.

0:09 – 0:50 · VÍCTOR TERRÓN-MACÍAS

5

Audit: can we trust the judge?

Mechanistic interpretability as a confidence proxy for LLM-assisted audits.

BLOCK 5 · AUDIT

Two tools you already have, one new subject: an LLM judge

> *“A plain probe can match an SAE feature, and patching tests what a model relies on.”*

Linear probe
reads what the hidden state represents

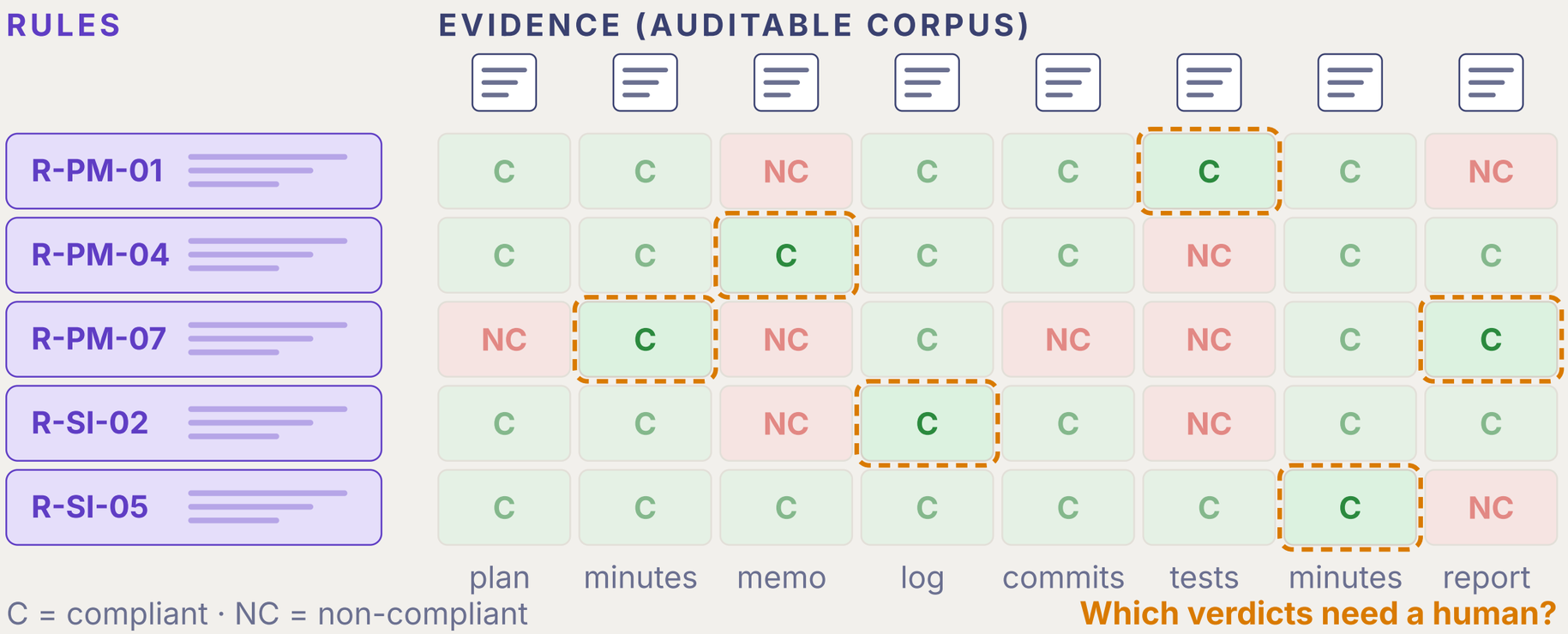
→ **Is this verdict likely wrong?**
a confidence signal read from inside, not from what the judge says

Intervention
here: editing the evidence text

→ **Does it rest on the right evidence?**
a grounding check for one verdict

BLOCK 5 · AUDIT

An LLM-assisted audit is a grid: every rule against every piece of evidence



Illustrative sizes. A real audit: tens of rules × hundreds of documents = thousands of verdicts.

BLOCK 5 · AUDIT · CONTEXT

Our motivating case: process audits of very small software teams

ISO/IEC 29110

Lifecycle profiles for **very small entities**: teams of up to 25 people. Covers project management and software implementation.

TR 29110-3-1

Guidelines for **assessing** those processes: check each expected practice against the project's own records.

> An assessor reads plans, minutes and change logs, rule by rule. That is the grid an LLM judge could help fill.

Paraphrased. The standard's text is licensed, so we never quote it; the rule on the next slides is invented.

BLOCK 5 · AUDIT · R-PM-07

Invented example, in the style of ISO/IEC 29110 · numbers illustrative

One rule, one piece of evidence: is the project compliant?

RULE R-PM-07

Every change request is 1 recorded, its 2 impact on schedule and cost is assessed, and 3 the customer approves it before work on it starts.

EVIDENCE · MEETING MINUTES

“12 Mar status meeting. Client asked to add CSV export to the reports module. Dev lead estimates 3 extra days; added to Sprint 4. Action: update the project plan.”

Invented example · numbers illustrative

The judge is confident — and wrong

LLM JUDGE

Compliant

“The request was recorded and its schedule impact assessed.”

Verbalized confidence  90%

HUMAN AUDITOR (GOLD)

Non-compliant

met	1 · recorded
missing	2 · cost impact not assessed
missing	3 · no customer approval before scheduling

90% confident · wrong

Invented example · numbers illustrative

A probe on the judge's residual stream disagrees, so a human gets the case



Invented example · numbers illustrative

Editing the evidence shows the verdict rests on one clue and ignores another

EVIDENCE · MEETING MINUTES

12 Mar status meeting. Client asked to add CSV export to the reports module. Dev lead estimates 3 extra days; added to Sprint 4. Action: update the project plan. **The customer rejected the change.**

JUDGE VERDICT

Compliant

Remove the schedule sentence	flips
Add a customer rejection	no change

Edit 2: schedule restored, rejection added

> The judge relies on the schedule clue and ignores the approval clause.

In one study, a judge's hidden states predicted its errors better than its words

Better calibrated

Brier-trained linear probes beat verbalized and multi-sample confidence, at about 10× less compute.

Transfers — there

On their reasoning judges, it held up on unseen domains. You will test that on ours.

Conservative

Under-confident on easy sets — a fair price when a missed error costs more than a review.

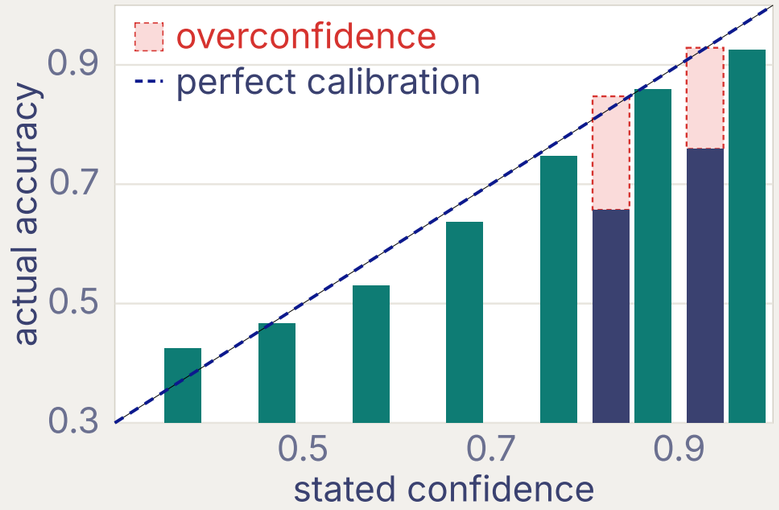
> LLM judges are overconfident in words. The cheap baseline every probe must beat: the judge's own Yes/No token probability.

Brier score = mean squared gap between confidence and the 0/1 outcome (lower is better). Radharapu et al., “Calibrating LLM Judges: Linear Probes for Fast and Reliable Uncertainty Estimation”, ACL 2026 Industry, arXiv 2512.22245. Related: semantic entropy probes, Kossen, Han et al. 2024, arXiv 2406.15927.

Synthetic illustration · not measured

A calibrated score turns review into a dial

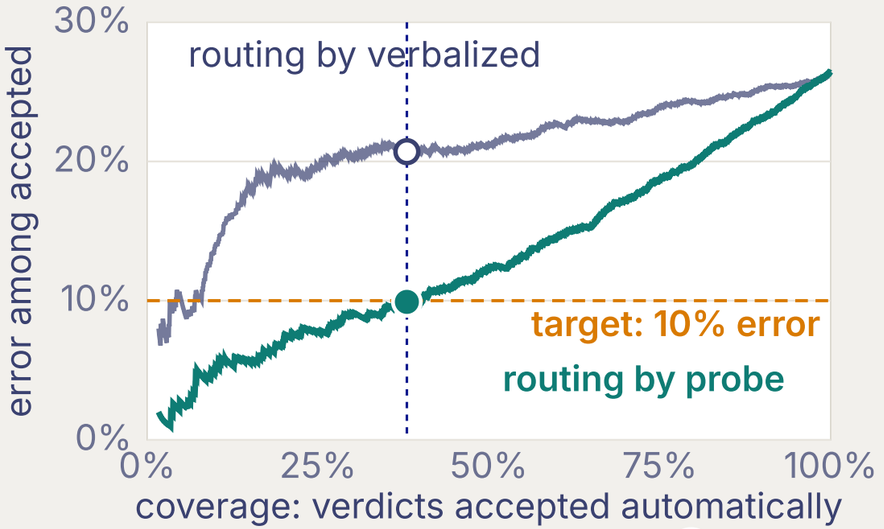
RELIABILITY DIAGRAM



Verbalized confidence · **ECE 0.17**

Probe on hidden state · **ECE 0.01**

RISK–COVERAGE



τ = 0.83

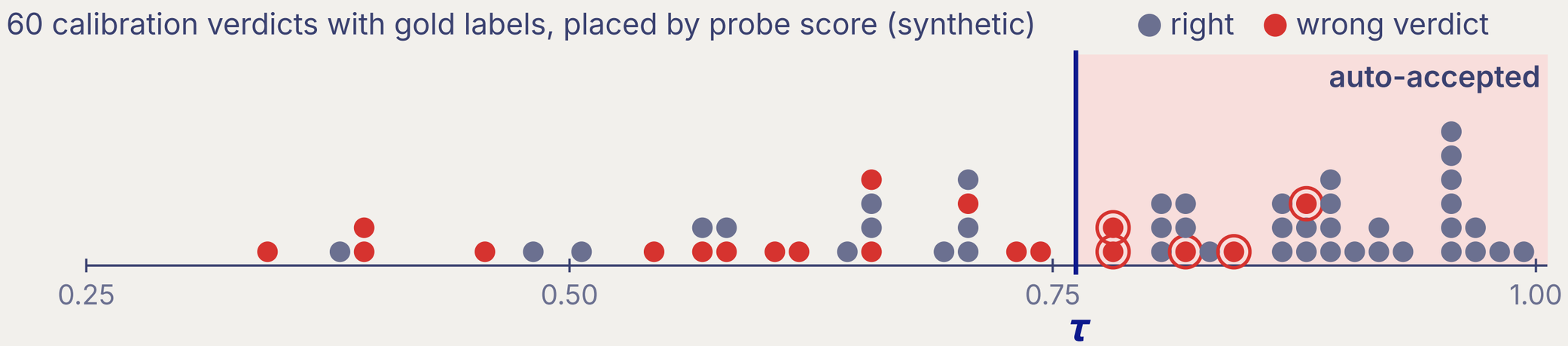
review threshold τ on probe score · drag or press →

auto-accept 38% · **62% to a human** · **3.8% escape**

Synthetic: 3,000 simulated verdicts. ECE = average gap between stated confidence and actual accuracy. Escape = wrong and auto-accepted, over all verdicts.

Research direction · not a published result

A conformal threshold could bound how many wrong verdicts escape review



α = 10% · **5 of 60 escape** (wrong and ≥ τ) · (60/61)·(5/60) + 1/61 = 9.8% ≤ 10% · auto-accept 34 of 60

Lowest τ with (n/(n+1))·escape(τ) + 1/(n+1) ≤ α on calibration data
→ expected escape rate ≤ α.

Holds only if new cases look like calibration cases. A new process breaks that: test it.

BLOCK 5 · AUDIT · HANDS-ON STAND-IN

In the notebook, contracts stand in for project records

AUDIT (R-PM-07)	LEGALBENCH CONTRACT-NLI
Rule	Assertion — one of 14 contract-NLI tasks
Evidence	Contract excerpt from an NDA
Verdict	Does this clause state that … ? Yes / No
A new 29110 process	3 held-out assertions (distribution shift)
Judge	Gemma 3 4B IT · cached hidden states, no GPU needed

LegalBench contract_nli_* tasks (CC BY 4.0): Guha et al., NeurIPS 2023 (Datasets & Benchmarks); derived from ContractNLI, Koreeda & Manning 2021.

NB5: measure whether the judge's hidden state knows when it is wrong

1. Run the judge on a few pairs
2. Train a probe to predict wrong verdicts
3. Compare with verbalized and token probability: AUROC, ECE
4. Pick a threshold: ≤ 10% error among accepted
5. Test on 3 held-out assertions
6. Remove the evidence behind confident verdicts

22

minutes, in pairs

NB5 judge confidence
Colab · CPU is fine
(cached activations)

AUROC chance a wrong verdict scores lower than a right one (0.5 = useless)
ECE average gap between stated confidence and actual accuracy
Brier mean squared gap between confidence and the 0/1 outcome

Our test run · Qwen2.5-0.5B fallback judge · not a published result

The judge's words are worse than chance. Its hidden state is not.

CONFIDENCE SIGNAL	AUROC	WHAT IT COSTS, AND WHAT IT IS
Verbalized — "how confident, 0–100?"	0.43	Worse than chance: it only ever answered 75, 80, 85 or 95.
Token probability — its own P(Yes/No)	0.63	Free, already in the logits. The baseline to beat.
Probe — linear read-out of the residual stream	0.71	One dot product, a few hundred labelled verdicts.

> Recalibration fixes ECE; it cannot fix AUROC. Ranking is where a probe has to win.

Judge accuracy 75 % on test. AUROC = chance a wrong verdict scores below a right one. NB5 sections 4–5.

Our test run · 3 assertions held out entirely

On a rule it has never seen, the probe stops being the better signal

64 % from 75 %

Judge accuracy on the three held-out assertions.

17.5 % vs 8 %

Verdicts escaping review at the probe's own threshold, against the free token-probability baseline.

What the probe may have learned

Not the judge's uncertainty, but *which rules are hard*.

And a correct verdict can be ungrounded

Of the 40 most confident *Yes* verdicts, 70 % stayed *Yes* on boilerplate evidence — and were **correct**.

NB5 sections 7–8. Per held-out assertion n = 40: read the direction, not the decimals.

BLOCK 5 · AUDIT · THE NUMBER YOU WOULD HAVE TO SIGN

A threshold fitted on a hundred verdicts inherits their noise

The policy

Auto-accept a verdict when its score ≥ `t`; otherwise it goes to a human. **Coverage** is the share you accept, **escape** the share that is wrong and accepted anyway.

NB5 takes the smallest `t` holding error under 10 % on calibration.

Why that is not a guarantee

On `cal` the target is met *by construction* — that is where the threshold was fitted.

On `test` it can be exceeded: tuned on roughly a hundred verdicts, it carries their sampling noise with it.

> Certify “at most 5 % of verdicts are wrong and unreviewed”. Is a threshold tuned on a hundred verdicts evidence for that?

NB5 section 6. A 5 % target left almost no coverage with this judge, hence 10 %.

BLOCK 5 · AUDIT · DEBRIEF

What would an auditor accept as evidence that a verdict is safe to skip?

Where did the probe win?

Against words, and against token probability?

Where did it fail?

Held-out assertions.
Ungrounded verdicts that are still correct.

What counts as evidence?

A score, a bound, a grounding check — or all three?

BLOCK 5 · AUDIT · METHOD MAP

Pick the method from your question, and the validation from the method

YOUR RESEARCH QUESTION	METHOD	VALIDATE WITH
Is concept X represented?	**Linear probe**	Held-out and OOD splits — BLOCK 5
Which component causes Y?	**Activation patching**	Other corruptions, path patching — BLOCK 5 · TEXT
Which features does it use?	**SAE / transcoder**	Beat a plain probe baseline
How is one prompt computed?	**Attribution graph**	Intervene on graph features
Can I control behaviour Z?	**Steering vector**	Try orthogonal directions

BLOCK 5 · AUDIT · YOUR TURN

Your turn: map one LLM-judged task from your own research

1
LLM-judged task
What does the model decide?

2
Ground truth
Who or what gives the gold label?

3
Confidence signal
Probe on which layer, which token?

4
Threshold policy
What escape rate can you tolerate?

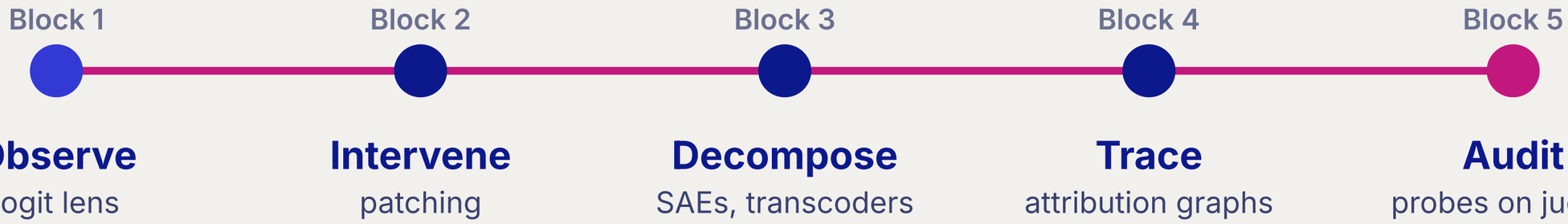
5
Grounding check
Which evidence should flip the verdict?

6
Shift test
What is your “unseen assertion”?

Two minutes in pairs in the tutorial · the transfer canvas is a printable in the repository

WRAP · WHERE THE FIELD IS GOING

From observing models to auditing them — with tools that are still young



Open problems

- Global circuits, not one prompt at a time
- Are replacement models faithful?
- Agents that do interpretability
- Shared evaluation standards

Implications

- Assurance and audits of AI systems
- Evidence for regulation
- Scientific discovery inside models

WRAP · RESOURCES

Everything we used today is open

TOOLS AND COURSES

TransformerLens
transformerlensorg.github.io/TransformerLens

SAELens · Gemma Scope 2
decoderesearch.github.io/SAELens

nnsight · circuit-tracer
nnsight.net · pypi.org/project/circuit-tracer

Neuronpedia
neuronpedia.org

Transformer Explainer
poloclub.github.io/transformer-explainer

ARENA · chapter 1
learn.arena.education

LegalBench · Guha et al. 2023
hazyresearch.stanford.edu/legalbench

PAPERS BEHIND TODAY

Interpretability in the Wild (IOI) · Wang et al. 2022
arxiv.org/abs/2211.00593

Toy Models of Superposition · Monosemanticity
transformer-circuits.pub (2022, 2023)

Circuit Tracing · Biology of an LLM
transformer-circuits.pub/2025/attribution-graphs

Are SAEs Useful? · Kantamneni et al. 2025
arxiv.org/abs/2502.16681

Open Problems · Sharkey et al., TMLR 2025
arxiv.org/abs/2501.16496

Calibrating LLM Judges · Radharapu et al. 2026
arxiv.org/abs/2512.22245

Gemma Scope 2 docs
ai.google.dev/gemma/docs/gemma_scope

Thank you

Miguel González
Tecnológico de Monterrey

Salvador Hinojosa
Tecnológico de Monterrey

Víctor Terrón-Macías
Tecnológico de Monterrey

Slides, notebooks and printables:

salvahin.github.io/mechinterp-tutorial